

ChimeraLM: a language model enables accurate structural variant detection in whole-genome amplified long-read sequencing

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Abstract

Single-cell genomic analysis relies on whole genome amplification (WGA) to generate sufficient DNA for sequencing, yet this process introduces chimera artifacts that manifest as false-positive structural variation (SV) and compromise downstream analyses. Here we present ChimeraLM, an interpretable genomic language model (GLM) that identifies WGA-induced chimera artifacts directly from sequence information. ChimeraLM is trained on matched WGA and bulk long-read sequencing from the same sample, using bulk support to label chimeric reads as amplification-induced artifacts or genuine genomic events. To capture long-range dependencies in variable-length reads, the model integrates Hyena operators with attention pooling. Evaluated on matched WGA and bulk nanopore datasets, ChimeraLM eliminated ~90% of chimeric reads from WGA data, restoring chimeric read proportion to near-bulk levels, whereas existing approaches achieved at most an 8% reduction. Benchmarked against a multi-platform gold-standard SV set, ChimeraLM removed 91–98% of unsupported SV calls from WGA datasets while retaining 75–87% of gold-standard SVs; against platform-matched bulk long-read call sets, it removed 91–99% of unsupported calls while retaining 75–86% of bulk-supported SVs. ChimeraLM further

normalized SV type distributions toward bulk profiles by suppressing the characteristic inversion bias observed in unprocessed WGA data. Attention-based interpretation indicates that ChimeraLM concentrates classification evidence at chimeric junctions, demonstrating its ability to learn biologically interpretable features. ChimeraLM provides a general approach for suppressing amplification-induced artifacts, enabling more reliable single-cell SV analysis across long-read platforms.

Keywords: Whole Genome Amplification, Single Cell, Genomic Language Model, Structural Variation

Main

Single-cell and low-input genomics have transformed our ability to resolve biological heterogeneity, enabling the discovery of rare cell states and the reconstruction of clonal evolution in cancer and development [1–3]. However, the limited DNA input (on the order of picograms per cell) makes comprehensive genome-wide profiling technically challenging [4, 5]. Whole genome amplification (WGA) therefore remains a prerequisite for high-coverage sequencing [6–8], yet it introduces systematic errors that compromise genomic fidelity, particularly for structural variation (SV) detection [9–11].

A major source of error in WGA is amplification-induced chimera formation. During this process, highly processive polymerases such as phi29, which is used in multiple displacement amplification (MDA), can switch templates and join discontinuous genomic loci into a single molecule [9–13]. As a result, WGA-based sequencing often produces chimeric reads that constitute a substantial fraction of the data [9]. These artificial sequences frequently create alignment patterns that closely resemble those generated by genuine SVs, including translocations and inversions [10]. Consequently, SV callers that rely on alignment-based signals (e.g., split-read and supplementary alignments) and coverage-derived evidence often misinterpret these amplification artifacts as true rearrangements, inflating false positives and distorting SV spectra [14–22]. This problem is particularly consequential for WGA-based long-read sequencing, which is otherwise well suited for resolving complex SVs at single-cell resolution [23, 24].

Distinguishing genuine genomic rearrangements from WGA-induced chimera artifacts remains a major computational challenge. Existing quality-control approaches typically rely on handcrafted rules or alignment-derived features, such as read orientation signatures or local coverage deviations [11, 13, 25]. However, these heuristics are often sensitive to platform- and protocol-specific variation. Moreover, they cannot capture sequence-level patterns or long-range dependencies within reads. As a result, low-input long-read sequencing remains difficult to deploy in settings where high precision is essential, including somatic mosaicism profiling [26] and validation of CRISPR off-target effects [27].

To address this challenge, we present ChimeraLM, an interpretable genomic language model (GLM) for identifying and filtering WGA-induced artifacts at the single-read level. Unlike existing approaches that rely on handcrafted rules derived

from read alignments or sequence-level features [11, 13, 25], ChimeraLM formulates artifact detection as a sequence-modeling task and learns discriminative features directly from raw reads [28]. Building on advances in DNA foundation models [29–32], it captures latent motifs and structural dependencies that generalize across long-read sequencing platforms provided by Oxford Nanopore Technologies (ONT). On ONT WGA datasets, ChimeraLM reduces chimeric reads by $\sim 90\%$ while preserving 75–87% of gold-standard SVs and 75–86% of platform-matched bulk-supported SVs, improving SV validation rates by 3.2- to 26.6-fold and mitigating WGA-induced biases in SV type distributions. Together, ChimeraLM provides an effective and interpretable filter for WGA long-read data, enabling robust SV discovery in single-cell and low-input genomics.

Results

Overview of ChimeraLM workflow and model architecture

ChimeraLM operates as a post-alignment filtering module within the single-cell long-read sequencing pipeline (Fig. 1a). After base calling and alignment to the reference genome, the resulting read set typically contains a mixture of true chimeric reads that arise from authentic genomic rearrangements and artificial chimeras introduced during WGA. ChimeraLM evaluates all reads with chimeric alignments before variant calling. For each read, the model determines whether the observed chimera reflects a genuine genomic event or a WGA-induced artifact. This binary classification enables selective removal of artificial chimeric reads while preserving true rearrangements, which improves the accuracy and sensitivity of downstream SV analyses.

To construct a robust supervised training set, we generated WGA long-read sequencing data from the human prostate cancer cell line PC3 using the ONT PromethION platform. For ground-truth calibration, we acquired three matched bulk long-read datasets from unamplified genomic DNA across diverse technologies: ONT PromethION, ONT MinION, and Pacific Biosciences (PacBio) Sequel II. Leveraging these references, we established a bulk-supported labeling framework by cross-referencing each WGA chimeric read against the chimeric alignment structures identified in the unamplified bulk data (Methods; Fig. 1b; Extended Data Fig. 1a). Under this classification scheme, WGA reads with alignment architectures corroborated by bulk evidence were labeled as genuine genomic events, while those lacking bulk support were classified as WGA-induced artifacts. To evaluate the model’s ability to generalize across sequencing hardware, we generated an independent WGA dataset on the MinION platform, which was reserved exclusively for testing (Extended Data Fig. 1a).

This labeling procedure classified 12,963,576 chimeric reads from the WGA PromethION dataset into two groups (genuine events and WGA-induced artifacts) based on bulk support (Extended Data Fig. 1b). Among these, 12,670,396 reads (97.7%) showed no matching alignment structures in any bulk dataset and were labeled as WGA-induced artifacts. The remaining 293,180 reads (2.3%) had matching structures in at least one bulk dataset, indicating they represent genuine genomic events rather than

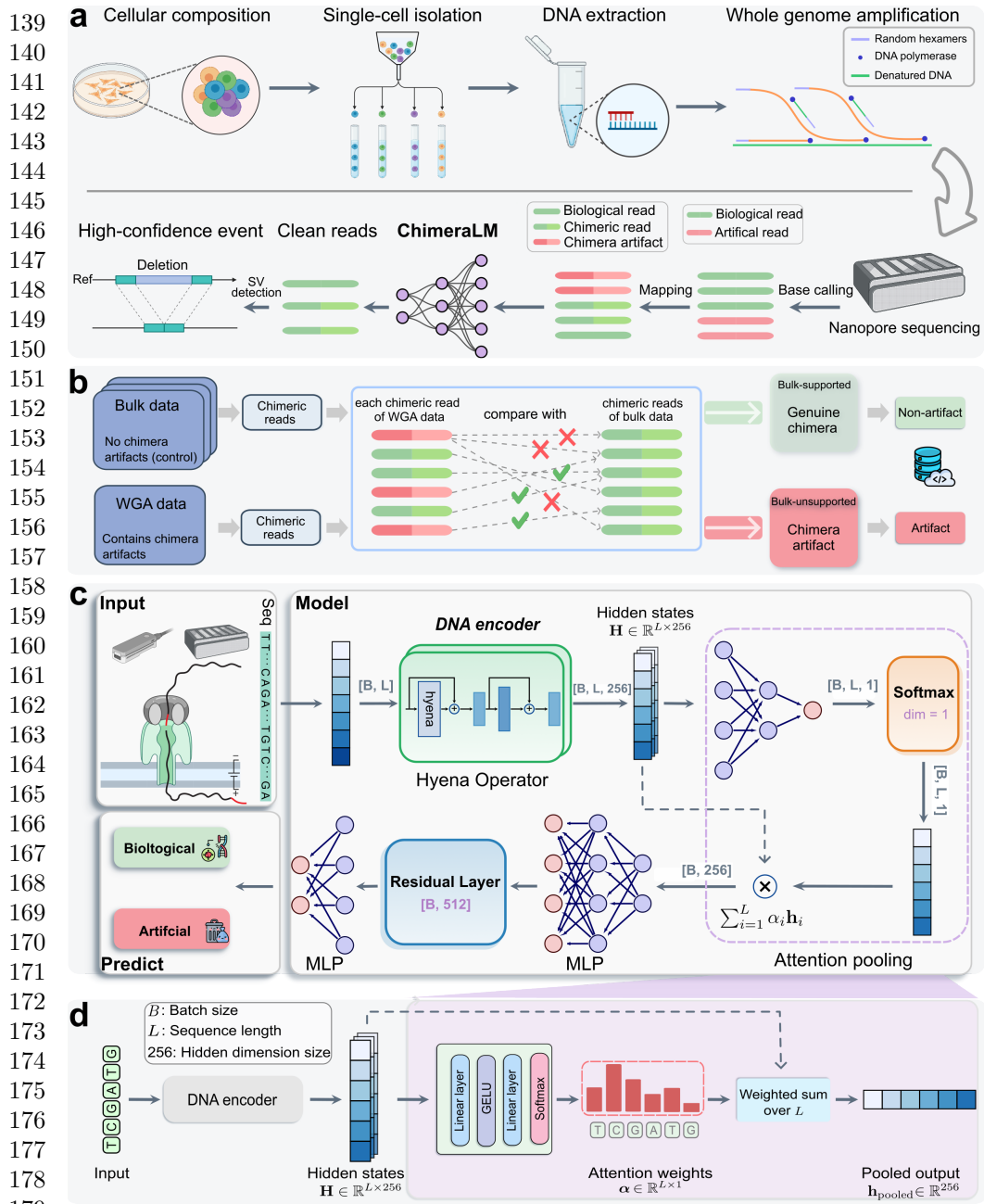


Fig. 1 ChimeraLM workflow and architecture for detecting WGA artifacts. (a) Single-cell genomic workflow and ChimeraLM integration. Single cells are isolated, followed by DNA extraction and WGA. During amplification, WGA-induced chimeric artifacts (red) are generated alongside genuine chimeric reads (green). After base calling and mapping, ChimeraLM classifies reads with chimeric alignments as genuine events or WGA-induced artifacts, enabling downstream SV detection on filtered data. (b) Bulk-supported label generation. Chimeric reads from WGA data are compared against bulk sequencing from the same cell line. Reads with bulk-supported alignment structures are labeled as genuine events (green); reads with no bulk match are labeled as WGA-induced artifacts (red). (c) ChimeraLM architecture. Input DNA sequences (batch size B , sequence length L) are tokenized at single-nucleotide resolution and encoded into hidden states $H \in \mathbb{R}^{L \times 256}$ through DNA encoder (HyenaDNA [29]). Hyena operators capture long-range dependencies. Attention pooling aggregates position-specific features, and multilayer perceptron (MLP) layers with residual connections process pooled representations for binary classification of genuine events and WGA-induced artifacts. (d) Attention pooling mechanism. Attention weights $\alpha \in \mathbb{R}^{L \times 1}$ are computed through linear layers with GELU activation and softmax normalization, assigning importance scores to each position. The weighted sum produces a fixed-dimensional representation $h_{\text{pooled}} \in \mathbb{R}^{256}$. Created with BioRender.com.

amplification artifacts, and were labeled as genuine chimeric reads. To construct a balanced training dataset, we retained all 293,180 genuine chimeric reads and randomly subsampled an equal number of WGA-induced artifacts. We further added 178,748 chimeric reads sampled from the bulk datasets to the genuine-event set, expanding the diversity of bulk-supported chimeric alignment structures used for training. The final labeled dataset comprised 765,108 reads and was split into training (70%), validation (20%), and test (10%) sets using stratified sampling (Extended Data Fig. 1a).

To model these labeled reads, ChimeraLM needs to process long, variable-length DNA sequences at single-nucleotide resolution (Fig. 1c). We therefore built ChimeraLM on HyenaDNA [29], a genomic foundation model pre-trained on diverse DNA sequences. Each read is tokenized at nucleotide resolution and encoded by Hyena operators [33], which capture long-range sequence context without splitting the input. The encoder produces a sequence of hidden states across the full read. To obtain a fixed-length representation for classification, ChimeraLM uses an attention-pooling module that learns position-specific weights and computes a weighted sum over the hidden states (Fig. 1d). The pooled representation is then passed through residual MLP blocks, and a final softmax outputs the probability that a read reflects a genuine event versus a WGA-induced artifact.

ChimeraLM achieves high accuracy and reduces artifacts to near-bulk levels across platforms

We first benchmarked ChimeraLM using the held-out test split (10%) derived from the bulk-supported labeled dataset (Extended Data Fig. 1a). This test set consists of chimeric reads labeled as genuine events or WGA-induced artifacts based on whether their chimeric alignment structures are supported by matched bulk sequencing. ChimeraLM achieved an F1 score of 0.81, with 0.95 recall and 0.70 precision (Fig. 2a). The high recall indicates that most WGA-induced artifacts are correctly identified for removal, which is important for limiting downstream false-positive SV calls, while the precision confirms that most reads flagged as artifacts are true amplification-induced chimeras rather than genuine genomic events.

Because ChimeraLM processes at most 32,768 bp per read, we asked how often this limit is reached and whether classification accuracy depends on read length (Extended Data Fig. 2). Chimeric reads in WGA data were predominantly short (median 2,100 bp on PromethION and 1,469 bp on MinION), and only 1,211 of 12,963,576 PromethION chimeric reads (0.009%) and 6 of 1,666,427 MinION chimeric reads exceeded 32,768 bp; in all but two of these reads, at least one chimeric junction lay within the first 32,768 bp (Extended Data Fig. 2a). Classification accuracy nevertheless varied with read length. On WGA-derived test reads, precision rose from 0.69 for reads ≤ 2 kb to ≥ 0.99 for reads > 8 kb, whereas recall declined from 0.98 (≤ 2 kb) to 0.76 (8–16 kb) and 0.31 (16–32 kb; $n = 80$) (Extended Data Fig. 2c). Long chimeric reads are therefore classified conservatively: errors favour retaining a read over removing it, and because reads longer than 16 kb constitute only 0.26% of PromethION chimeric reads, this length dependence has little effect on aggregate performance. This behaviour likely reflects the composition of the training data, in which chimeric reads sampled from bulk sequencing (median 14.4 kb in the test split) are far longer than WGA-derived

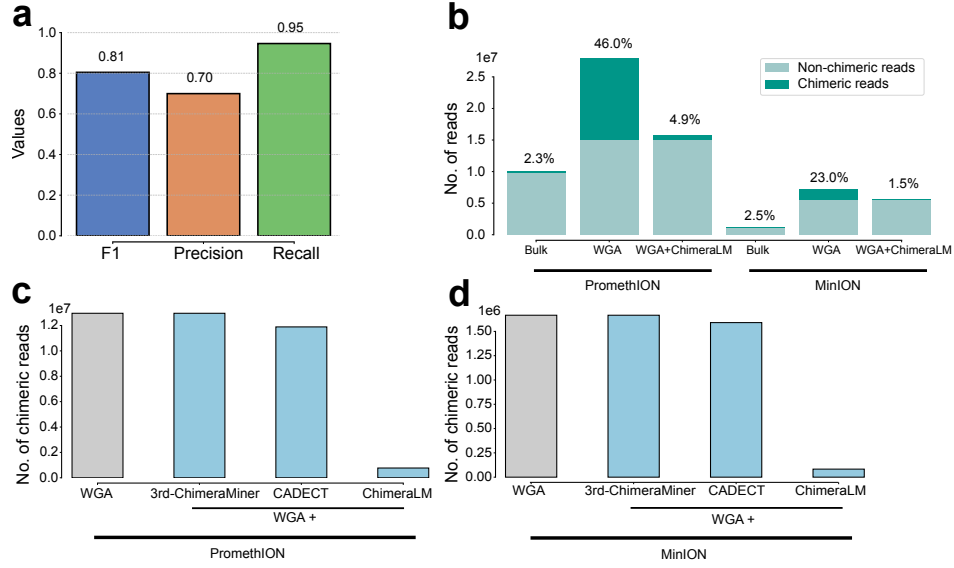


Fig. 2 ChimeraLM accurately identifies and removes WGA-induced chimeric artifacts. (a) Classification performance on held-out test data. ChimeraLM achieves recall of 0.95, precision of 0.70, and F1 score of 0.81. (b) Chimeric read reduction across sequencing platforms. Stacked bars show proportions of chimeric (dark teal) and non-chimeric (light teal) reads in bulk, WGA, and ChimeraLM-filtered samples. ChimeraLM reduces chimeric read frequencies from 46.0% to 4.9% (PromethION) and from 23.0% to 1.5% (MinION), approaching bulk levels (2.3% and 2.5%, respectively). (c,d) Benchmarking against existing methods on PromethION (c) and MinION (d). The gray bar indicates the total number of chimeric read on unfiltered WGA data. The blue bar represents the total number of chimeric reads remaining after filtering by each method. ChimeraLM achieves approximately 90% reduction in chimeric reads on both platforms, while 3rd-ChimeraMiner shows no detectable reduction and CADECT shows 8.3% and 4.6% reduction on PromethION and MinION, respectively. SACRA failed to complete due to memory exhaustion (> 500 GB RAM required).

reads (median 1.5–2.1 kb), so that read length itself carries label information. Accordingly, the chimeric reads removed by ChimeraLM were shorter than those retained (PromethION: median 2,067 versus 3,087 bp, N50 3,836 versus 7,989 bp; MinION: median 1,458 versus 2,031 bp, N50 2,220 versus 5,560 bp; Extended Data Fig. 2d), even though among WGA-derived test reads bulk-supported chimeras were slightly shorter than artifacts (median 1,459 versus 2,086 bp). Filtering thus shifts the chimeric-read length distribution modestly upwards and under-removes rare long artifacts, whereas the length distribution of non-chimeric reads, which make up the majority of a WGA library and are not evaluated by ChimeraLM, is unchanged.

We next examined whether applying ChimeraLM filtering would restore chimeric read rates in full PC3 WGA datasets toward the levels observed in bulk sequencing across both PromethION and MinION platforms (Fig. 2b). Bulk sequencing established low baseline chimeric read rates of 2.3% (PromethION) and 2.5% (MinION). In contrast, WGA increased the chimeric read fraction to 46.0% and 23.0%, respectively. After ChimeraLM filtering, chimeric content dropped to 4.9% on PromethION and 1.5% on MinION, corresponding to 10- to 15-fold reductions, while retaining 15.8

million and 5.6 million reads. These post-filtering rates approach the corresponding bulk baselines, indicating effective removal of WGA-induced artifacts while preserving genuine signal.

We compared ChimeraLM against SACRA [25], 3rd-ChimeraMiner [13], and CADECT [11], existing tools for detecting amplification-induced chimeras (Fig. 2c,d). ChimeraLM reduced chimeric reads by ~90% on both platforms, outperforming CADECT (8.3% and 4.6% reduction on PromethION and MinION, respectively), while 3rd-ChimeraMiner showed no detectable reduction. SACRA could not be evaluated due to out-of-memory errors even with 500 GB RAM.

The MinION results further provide an independent test of model generalization, as this MinION WGA dataset was not used during training. ChimeraLM was trained exclusively on PromethION WGA data, yet achieved comparable chimeric read reduction on MinION. This cross-platform generalization indicates that ChimeraLM captures sequence-level features intrinsic to WGA-induced artifacts rather than platform-specific signatures, supporting its potential applicability to additional long-read and potentially short-read sequencing technologies.

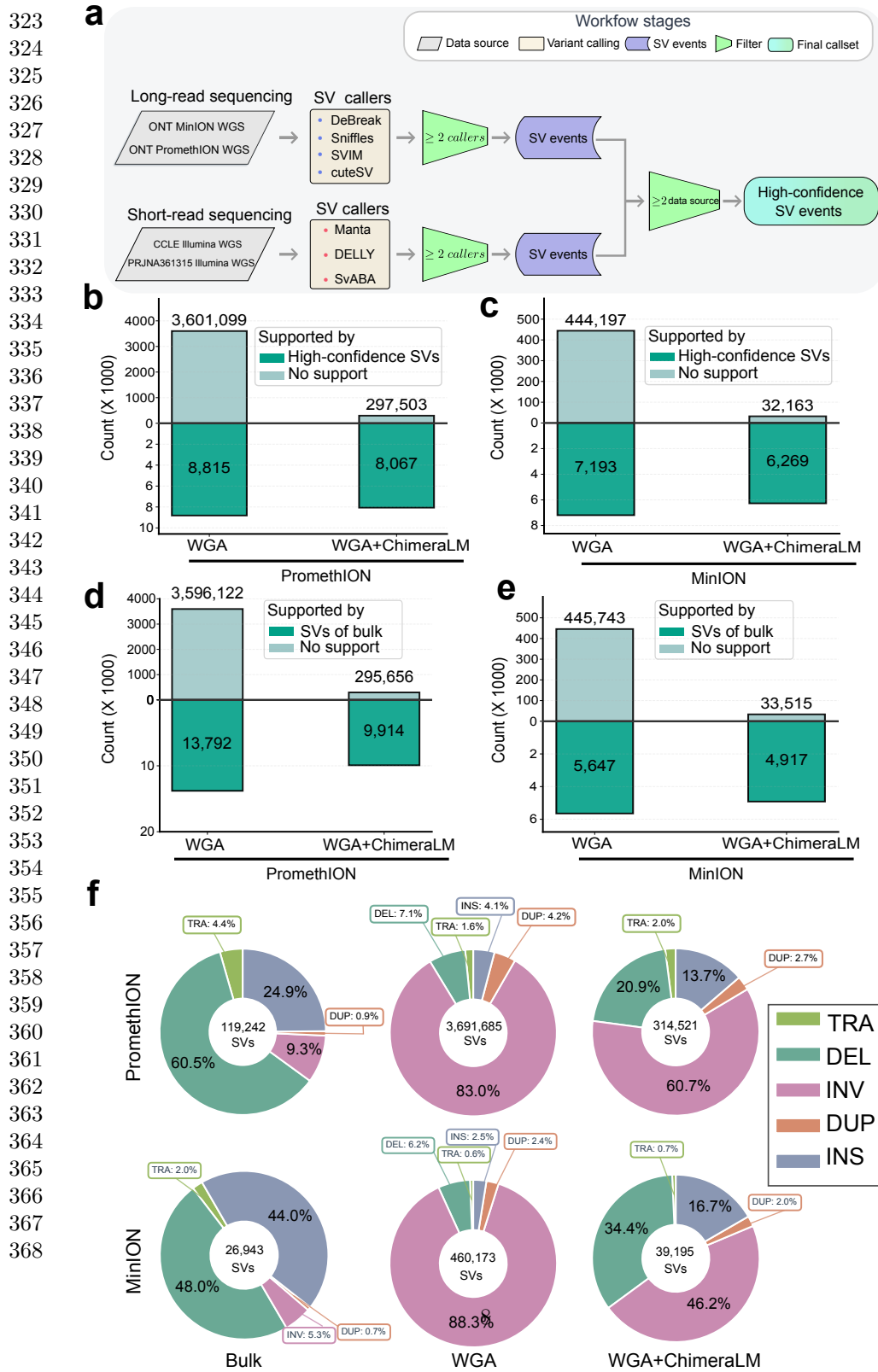


Fig. 3 ChimeraLM improves structural variant detection accuracy. (a) Construction of a high-confidence SV reference dataset from bulk PC3 sequencing. Four bulk datasets were integrated: ONT MinION Mk1C, ONT PromethION P2, the CCLE Illumina whole genome sequencing (WGS) dataset, and the PRJNA361315 Illumina WGS dataset. SVs were called independently within each dataset using multiple callers, and events supported by ≥ 2 callers per dataset were retained. SVs were then compared across datasets, and events observed in ≥ 2 of the four bulk datasets were designated as gold-standard SVs. (b,c) SV validation against the gold-standard reference for PromethION (b) and MinION (c). Bars show SV calls supported by the gold standard (dark teal) or unsupported (light teal). (d,e) SV validation against platform-matched long-read bulk sequencing for PromethION (d) and MinION (e), capturing true long-read SVs that may not be represented in the multi-platform reference. Bars show SV calls supported by the platform-matched long-read bulk data (dark teal) or unsupported (light teal). (f) Distribution of SV types for PromethION and MinION datasets across bulk samples, unfiltered WGA, and WGA with ChimeraLM filtering. Unfiltered WGA exhibits an excess of inversion (INV) events, which is substantially reduced following ChimeraLM filtering. Other SV classes, including translocation (TRA), deletion (DEL), duplication (DUP), and insertion (INS), are also shown.

ChimeraLM reduces unsupported structural variant calls

Accurate SV detection from single cells is essential for characterizing genomic diversity and disease mechanisms. However, WGA-induced chimeric artifacts can be misinterpreted as genuine SVs, potentially leading to incorrect biological inferences. To assess the impact of ChimeraLM on SV calling, we compared SV callsets generated from unfiltered WGA reads with those obtained after applying ChimeraLM filtering (WGA + ChimeraLM). Both callsets were evaluated against two complementary bulk-derived references: (i) a stringent gold-standard SV set derived from bulk PC3 DNA through cross-dataset consensus, and (ii) platform-matched long-read bulk SV callsets used as platform-specific references.

We first constructed a high-confidence gold-standard SV set from bulk PC3 DNA by integrating four independent sequencing datasets: ONT PromethION, ONT MinION, and two Illumina short-read WGS datasets from the Cancer Cell Line Encyclopedia (CCLE Illumina WGS [34]) and from a previously published study (PRJNA361315 Illumina WGS [35]) (Fig. 3a; Extended Data Table 1). SVs were called separately within each dataset using multiple SV callers. Events supported by at least two callers within a dataset were retained, and only SVs observed in at least two of the four datasets were designated as gold-standard events.

Relative to this stringent gold standard, unfiltered WGA produced a large number of unsupported SVs. On PromethION, WGA yielded 3,601,099 SV calls, of which only 8,815 (0.24%) overlapped gold-standard events. After ChimeraLM filtering, total calls decreased to 305,570 (a 91.5% reduction) while retaining 8,067 gold-standard events (91.5% retention), increasing the validation rate to 2.64% (11-fold) (Fig. 3b). On MinION, total calls decreased from 451,390 to 38,432 (a 91.5% reduction), while gold-standard-supported events decreased from 7,193 to 6,269, corresponding to 87.2% retention. The validation rate increased from 1.59% to 16.3% (10-fold) (Fig. 3c).

Because the gold standard is intentionally stringent and may exclude true SVs detectable only in long-read data, we next performed platform-matched validation using long-read bulk sequencing from the same platform (Fig. 3d,e). This analysis provides a platform-specific estimate of recall and reduces bias introduced by the strict gold-standard definition. ChimeraLM increased validation rates from 0.38% to 3.24% on PromethION (8.5-fold) and from 1.25% to 12.79% on MinION (10-fold), while retaining 71.9% and 87.1% of bulk-supported events, respectively. Together, these results show that ChimeraLM removes the vast majority of unsupported SV calls while preserving the majority of bulk-supported variants across platforms.

ChimeraLM mitigates WGA-induced SV type biases

Amplification artifacts can profoundly distort the apparent spectrum of SVs, complicating biological interpretation. To quantify these effects, we compared SV-type compositions across bulk, unfiltered WGA, and ChimeraLM-filtered datasets generated on both ONT PromethION and MinION platforms (Fig. 3f). Bulk datasets displayed SV profiles dominated by deletions and insertions, whereas WGA datasets exhibited a striking inflation of inversions, reaching 83.0% on PromethION and 88.3% on MinION. This inversion-heavy signature mirrors previous observations from

single-cell long-read sequencing using PacBio and droplet-based MDA [22], suggesting that many of these events arise from amplification-induced chimeras rather than genuine genomic rearrangements.

To directly assess which SV types are preferentially associated with amplification artifacts, we examined SV calls supported exclusively by reads classified as WGA-derived chimeras. These artifact-supported events were overwhelmingly enriched for inversions, comprising 88.4% on PromethION and 92.4% on MinION (Extended Data Fig. 3). Smaller fractions of deletions, duplications, and insertions were also present, indicating that WGA artifacts inflate not only inversion calls but generate false positives across multiple SV categories.

Applying ChimeraLM substantially reduced these biases. The inversion fraction decreased from 83.0% to 60.7% on PromethION and from 88.3% to 46.2% on MinION, while the relative abundance of deletions and insertions increased toward bulk-like levels (Fig. 3f). By selectively removing artifact-supported events, ChimeraLM suppresses the dominant inversion inflation and reduces spurious calls across SV types, shifting SV-type distributions toward bulk profiles and improving the accuracy and interpretability of single-cell SV analyses [22].

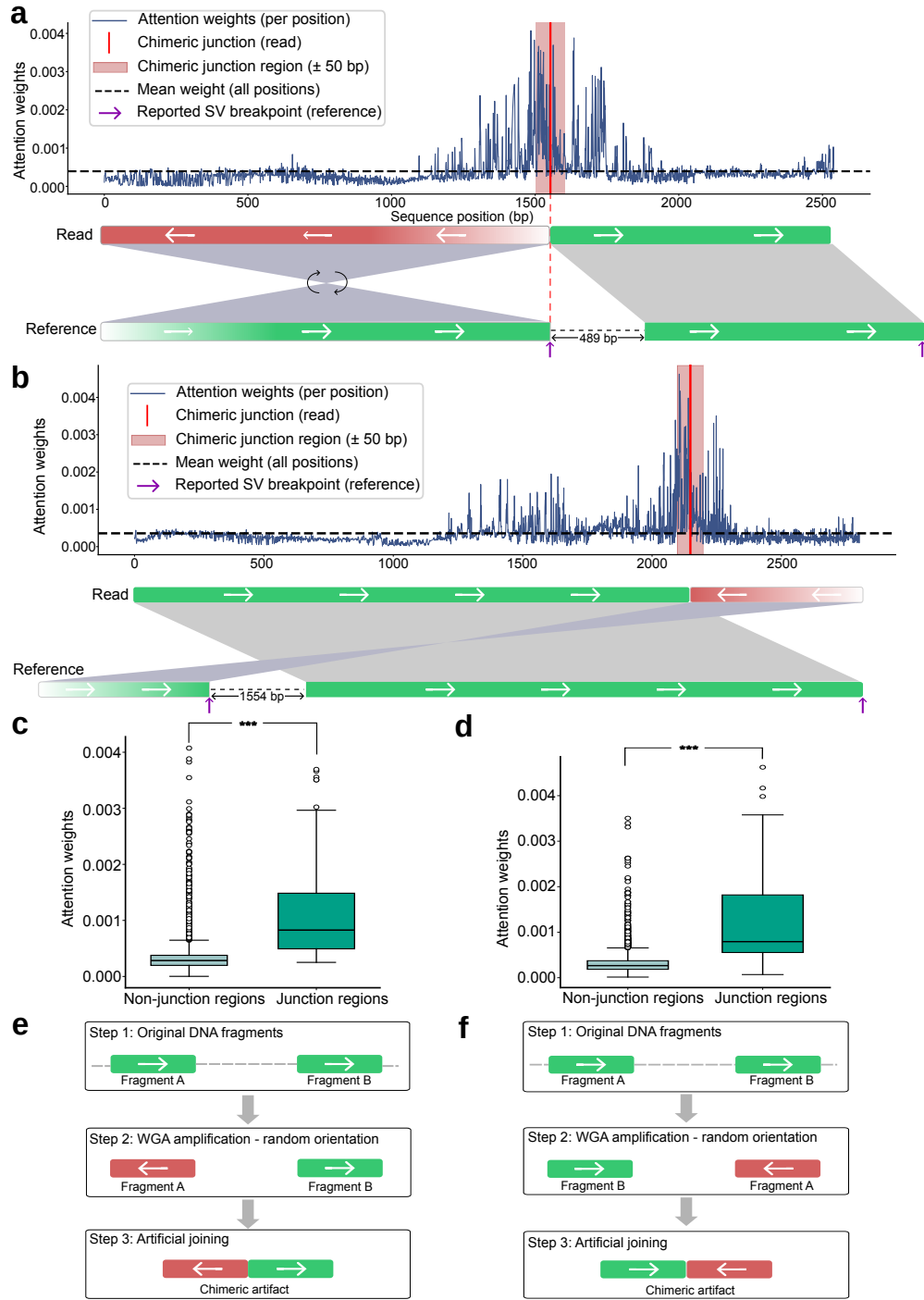


Fig. 4 ChimeraLM attention weights are enriched at chimeric junction regions. (a,b) Attention weight profiles for two representative chimeric reads exhibiting distinct junction configurations. Upper panels show per-position attention weights (blue) with the mean attention across the read indicated by a dashed line. Red vertical lines mark inferred chimeric junction positions, and pink shading denotes the junction-centered region (± 50 bp). Lower panels display read-level alignments, highlighting orientation transitions at the junctions (green, forward orientation; red, reverse-complemented orientation). (c,d) Quantitative comparison of attention weights between junction and non-junction regions. Junction-centered windows show significantly elevated attention weights relative to non-junction regions ($P = 5.3 \times 10^{-14}$ and $P = 6.8 \times 10^{-15}$; Wilcoxon rank-sum test). (e,f) Schematic illustration of WGA-induced chimera formation. During amplification, DNA fragments originating from distant genomic loci can be amplified in either orientation, joining them into a single molecule with discordant orientations, producing inversion-like alignment signatures. The two examples illustrate forward-to-reverse and reverse-to-forward orientation transitions.

Attention visualization reveals interpretable classification features

We examined whether ChimeraLM’s attention weights provide an interpretable signal by focusing on mechanistically relevant regions of chimeric reads, particularly the junctions during WGA (Fig. 4). We inspected two representative chimeric reads exhibiting distinct junction configurations (Fig. 4a,b). In both cases, attention remained relatively flat across most positions but formed sharp, concentrated peaks at the inferred junction regions. These peaks aligned with the read-level breakpoint separating two genomic loci and coincided with an orientation transition between adjacent alignment segments.

To quantify this effect, we compared attention weights within junction-centered windows (± 50 bp) against weights from non-junction regions (Fig. 4c,d). Junction windows showed significantly higher attention weights (Wilcoxon rank-sum test, $P = 5.3 \times 10^{-14}$ and $P = 6.8 \times 10^{-15}$), indicating that ChimeraLM preferentially emphasizes sequence context proximal to chimeric junctions.

This attention enrichment is consistent with the expected structure of WGA-induced chimeras (Fig. 4e,f): DNA fragments from distant loci can be amplified in either orientation and subsequently joined, generating junctions with discordant orientations. Together, these results suggest that ChimeraLM’s attention peaks provide a mechanistically interpretable signal that concentrates classification evidence to junction-proximal sequence positions within individual reads.

Discussion

WGA enables genomic analysis from single cells and other low-input samples, but it also introduces chimeric artifacts that compromise SV detection. ChimeraLM addresses this challenge by classifying chimeric reads as genuine events or WGA-induced artifacts directly from read information, and removing artifacts before variant calling, rather than attempting to correct artifact-driven calls post hoc. Across ONT platforms, ChimeraLM improved data quality at the read and variant levels: it reduced chimeric reads by $\sim 90\%$ while retaining 75–87% of gold-standard SVs and 75–86% of platform-matched bulk-supported SVs, and it increased the fraction of supported SV calls by 3.2- to 26.6-fold. Notably, performance generalized from PromethION (used for training) to MinION without platform-specific retraining, suggesting that ChimeraLM captures properties shared by WGA-induced artifacts rather than instrument-specific signatures.

In comparison, existing methods showed limited effectiveness on our ONT WGA datasets. Two tools originally developed and primarily evaluated on PacBio data (SACRA and 3rd-ChimeraMiner) [13, 25] either failed to complete under our evaluation setting (SACRA, > 500 GB RAM) or showed no detectable reduction (3rd-ChimeraMiner), highlighting poor cross-platform generalization. CADECT [11], which was designed for ONT data, achieved only 8.3% and 4.6% reduction on PromethION and MinION, respectively. CADECT detects concatemers via sliding-window self-alignment, an effective strategy for repeat-like structures with internal sequence similarity, but it is not designed to capture the broader diversity of WGA-induced

chimeras. Together, these comparisons suggest that rule-based or subtype-specific heuristics do not comprehensively address amplification-induced chimeras and motivate learning-based models that can extract discriminative sequence signatures without imposing predefined structural assumptions.

ChimeraLM also illustrates the value of deep learning for quality-control problems where conventional alignment- and coverage-derived criteria provide limited resolution. [11, 13, 25, 28]. By learning directly from sequence, ChimeraLM discovers subtle compositional and structural features that separate genuine events from amplification artifacts. The model also offers interpretability through attention visualization: attention weights concentrate at junction regions where template switching joins discordant loci, validating the biological relevance of the learned features. These methodological advances have direct implications for single-cell genomics, where high false-positive rates in WGA data have constrained robust characterization of chromosomal instability, clonal evolution, and SV burden [20, 22, 36]. By improving the signal-to-noise ratio and clarifying SV-type spectra that are otherwise distorted by amplification artifacts, ChimeraLM enables more confident identification of genuine SVs, supporting studies of cancer evolution, developmental biology, and somatic mosaicism where single-cell resolution is essential [26, 27].

Several limitations warrant consideration. First, the current model processes reads independently; integrating contextual features such as coverage or phasing information may further enhance accuracy. Second, regarding computational resources, while central processing unit (CPU) inference is feasible, graphics processing unit (GPU) acceleration is recommended for processing large-scale datasets. Finally, future work should extend validation to diverse cell types, sequencing platforms (e.g., PacBio HiFi), and alternative WGA protocols, including multiple annealing and looping-based amplification cycles (MALBAC) [37], linear amplification via transposon insertion (LIANTI) [5], primary template-directed amplification (PTA) [19], and droplet-based MDA (dMDA) [38]. While the sequence-based approach suggests broad applicability, systematic validation across amplification chemistries is needed to assess generalization limits and optimize performance for specific protocols.

More broadly, ChimeraLM illustrates the potential of GLMs for data quality control. As long-context architectures continue to advance [29], extending the model’s context window to handle megabase-scale inputs could enable artifact detection in more complex genomic structures. This framework could extend to other amplification-dependent technologies, such as cell-free DNA analysis, ancient DNA studies, and low-biomass metagenomics. Furthermore, attention-based interpretability opens opportunities for studying template-switching dynamics, potentially guiding the development of improved amplification protocols. In summary, ChimeraLM provides a practical and interpretable framework for enhancing long-read single-cell genomic fidelity, ensuring that downstream biological insights are derived from genuine SVs rather than technical artifacts.

599 Methods

600 Cell culture, single-clone preparation, and nanopore sequencing

602 *Cell culture and single-clone establishment*

603 PC3 prostate cancer cells (ATCC[®] CRL-1435[™]) were cultured in RPMI-1640 medium
604 supplemented with 10% fetal bovine serum and 1% penicillin–streptomycin at 37 °C
605 with 5% CO₂. To minimize biological heterogeneity, a monoclonal population was
606 established by serial dilution in 96-well plates, ensuring that each culture originated
607 from a single cell. Mycoplasma contamination was routinely tested and confirmed
608 negative prior to DNA extraction.

610 *DNA extraction and whole-genome amplification*

611 From the monoclonal population, two types of DNA samples were prepared: a
612 bulk (non-amplified) control and ten single-cell MDA-amplified genomes. Bulk high-
613 molecular-weight DNA was extracted using the Monarch[®] HMW DNA Extraction
614 Kit for Cells & Blood (New England Biolabs). Individual cells were isolated using
615 1CellDish-60 mm (iBiochips) and amplified using the REPLI-g Advanced DNA Sin-
616 gle Cell Kit (Qiagen) following the manufacturer’s protocol. DNA concentration and
617 fragment integrity were assessed with a Qubit 4 fluorometer and Agilent TapeStation
618 (DNA 1000/5000 ScreenTape). Only samples meeting quality standards were used for
619 library construction.

621 *Nanopore library preparation and sequencing*

622 Libraries were prepared using the [ONT](#) Ligation Sequencing Kit V14 (SQK-LSK114)
623 and sequenced on MinION Mk1C or PromethION P2 Solo devices with R10.4.1 flow
624 cells following the manufacturer’s genomic DNA workflow. Because all single-cell sam-
625 ples originated from the same monoclonal lineage, differences between amplified and
626 bulk datasets primarily reflect MDA-induced artifacts rather than biological variation.

628 *Basecalling and read processing*

629 POD5 files were basecalled using Dorado v0.5.0 with the high-accuracy model
630 `dna_r10.4.1_e8.2.400bps_hac@v4.3.0` [39]. Reads with mean quality < 10 or
631 length < 500 bp were removed. Adapters and concatemers were trimmed using
632 Cutadapt v4.0 [40] in a two-pass, error-tolerant procedure. Filtered reads were aligned
633 to the GRCh38.p13 reference genome using minimap2 v2.26 (`map-ont` preset) [41].
634 BAM files were sorted and indexed using SAMtools v1.16 [42]. Read-length and map-
635 ping statistics were computed using NanoPlot v1.46.1 [43]. All samples were processed
636 using identical parameters.

638 *Chimeric read identification*

639 Chimeric reads were identified from BAM files using [supplementary alignment \(SA\)](#)
640 tags. Reads were classified as chimeric if they (i) were mapped, (ii) contained an SA
641 tag, (iii) were primary alignments (not secondary), and (iv) were not supplementary
642 alignments themselves. This definition counts each chimeric read once using its primary
643 alignment.

alignment while excluding secondary/supplementary records, thereby avoiding double-counting and reducing ambiguity from low-confidence alignments. Reads lacking SA tags were classified as non-chimeric.

Training data construction

Data generation and sources

To construct the training dataset, we generated WGA and bulk sequencing data from PC3 cells. The WGA sample was amplified and sequenced on the PromethION P2 platform (ONT), while three independent bulk datasets were produced from non-amplified genomic DNA: bulk PromethION P2, bulk MinION Mk1c (ONT), and bulk PacBio. These bulk datasets represent authentic biological sequences free from amplification-induced artifacts. In contrast, WGA sequencing includes both genuine genomic reads and artificial chimeras introduced during the amplification process.

Ground truth annotation and class definition

Ground truth labels were established by systematically comparing chimeric reads from the WGA PromethION P2 dataset against those from the three bulk datasets. For each WGA chimeric read, all alignment segments—defined by their genomic start and end coordinates—were compared to the corresponding segments of bulk chimeric reads. A WGA read was labeled as biological if every segment matched at least one bulk chimeric read within a 1 kb positional tolerance, indicating that the structural configuration is also present in non-amplified DNA. Reads lacking any matching pattern across all bulk datasets were labeled as artificial chimeras, presumed to arise from the amplification process. Additional chimeric reads were randomly sampled from the bulk datasets and labeled as biological, as these reads originate from genuine genomic rearrangements such as true SVs. The final labeled dataset combined the annotated WGA PromethION P2 reads with the subsampled bulk chimeric reads and was subsequently partitioned into training, validation, and test sets as described below.

Dataset partitioning and cross-platform validation

The combined labeled dataset, derived from WGA PromethION P2 and bulk sequencing data, was divided into training (70%), validation (20%), and test (10%) sets using stratified random sampling. These subsets were used respectively for model training, hyperparameter tuning, and performance evaluation on data from the same sequencing platform.

To evaluate cross-platform generalization, the complete WGA MinION Mk1c dataset was reserved. This dataset, generated on a different nanopore platform, was never used during model training or internal testing. This two-level evaluation design allowed us to test whether ChimeraLM captures general sequence features of amplification-induced chimeras rather than platform-specific artifacts.

691 Model architecture

692 *DNA encoder*

693 ChimeraLM employs the pre-trained HyenaDNA model [29] as its DNA encoder. This
694 model was pre-trained on large-scale genomic data and provides robust sequence rep-
695 resentations. DNA sequences are tokenized at single-nucleotide resolution, with each
696 base (A, C, G, T, N) mapped to a unique integer token (7, 8, 9, 10, 11, respectively).
697 Special tokens include [CLS]=0, [PAD]=4, and others for sequence processing. Input
698 sequences are truncated at 32,768 bp or padded to enable batch processing; reads
699 longer than 32,768 bp are not discarded but are classified from their first 32,768 bp
700 (Extended Data Fig. 2).

701 For a tokenized input sequence $\mathbf{x} \in \mathbb{Z}^L$, the HyenaDNA generates contextualized
702 hidden representations:

$$703 \mathbf{H} = \text{HyenaDNA}(\mathbf{x}) \in \mathbb{R}^{L \times 256}$$

704 where $\mathbf{H} = (\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_L)$ represents position-wise hidden states with dimension 256.
705 The Hyena operators [33] efficiently capture both local sequence motifs and long-range
706 dependencies essential for distinguishing biological sequences from chimeric artifacts.

707 *Attention pooling*

708 To aggregate variable-length sequence representations into fixed-size vectors,
709 ChimeraLM implements attention-based pooling. For hidden states $\mathbf{H} \in \mathbb{R}^{L \times 256}$,
710 attention weights are computed through a two-layer network:

$$711 \mathbf{e} = \text{GELU}(\text{Linear}_{256 \rightarrow 256}(\mathbf{H})) \in \mathbb{R}^{L \times 256}$$

$$712 \mathbf{s} = \text{Linear}_{256 \rightarrow 1}(\mathbf{e}) \in \mathbb{R}^{L \times 1}$$

$$713 \boldsymbol{\alpha} = \text{softmax}(\mathbf{s}) \in \mathbb{R}^{L \times 1}$$

714 The pooled representation is the weighted sum of hidden states:

$$715 \mathbf{h}_{\text{pooled}} = \sum_{i=1}^L \alpha_i \mathbf{h}_i \in \mathbb{R}^{256}$$

716 This mechanism assigns learned importance weights to each sequence position,
717 enabling the model to focus on informative regions while accommodating natural
718 variability in read lengths.

719 *Classification head*

720 The pooled representation is processed through a MLP with residual connections. The
721 first layer expands dimensionality:

$$722 \mathbf{f}_1 = \text{Dropout}_{0.1}(\text{GELU}(\text{Linear}_{256 \rightarrow 512}(\mathbf{h}_{\text{pooled}}))) \in \mathbb{R}^{512}$$

Subsequent residual blocks with input $\mathbf{f}_{\text{in}} \in \mathbb{R}^{512}$ compute: 737

$$\mathbf{f}_{\text{out}} = \text{Dropout}_{0.1}(\text{Linear}_{512 \rightarrow 512}(\text{GELU}(\text{Linear}_{512 \rightarrow 512}(\mathbf{f}_{\text{in}})))) + \mathbf{f}_{\text{in}} \quad 738$$

where the skip connection enables stable gradient flow during training. The final layer produces binary classification logits: 739

$$\mathbf{z} = [z_0, z_1] = \text{Linear}_{512 \rightarrow 2}(\mathbf{f}_{\text{final}}) \in \mathbb{R}^2 \quad 740$$

where z_0 and z_1 represent logits for biological and artificial chimeric classes, respectively. During inference, the predicted class is $\hat{y} = \text{argmax}_{i \in \{0,1\}} z_i$. 741

Model summary 742

The complete ChimeraLM pipeline processes DNA sequences through: (1) single-nucleotide tokenization, (2) HyenaDNA backbone encoding to generate contextualized representations, (3) attention pooling to aggregate position-specific features, (4) MLP layers with residual connections to learn classification features, and (5) binary classification output. The entire model with $\sim 4.2\text{M}$ trainable parameters is trained end-to-end using labeled data. 743

Model training and optimization 744

Training configuration 745

ChimeraLM was trained using PyTorch [44] and PyTorch Lightning [45] frameworks. Input sequences were tokenized using the tokenizer with maximum sequence length of 32,768 bp. Sequences longer than this threshold were truncated; shorter sequences were padded to enable batch processing. Training employed mixed-precision computation (bf16) to accelerate training while maintaining numerical stability. 746

Optimization procedure 747

We used the AdamW optimizer [46] with learning rate $\eta = 1 \times 10^{-4}$ and weight decay $\lambda = 0.01$. AdamW implements adaptive learning rates with decoupled weight decay, combining the benefits of Adam optimization with proper L2 regularization. A ReduceLROnPlateau scheduler dynamically adjusted the learning rate based on validation loss, reducing it by a factor of 0.1 when no improvement occurred for 10 consecutive epochs. Early stopping with patience of 10 epochs prevented overfitting by terminating training when validation performance plateaued. A fixed random seed (12345) ensured reproducibility across training runs. 748

The training objective used cross-entropy loss for binary classification. For a training example with class label $y \in \{0, 1\}$ and model logits $\mathbf{z} = [z_0, z_1]$, the loss is: 749

$$\mathcal{L}(\mathbf{z}, y) = -\log\left(\frac{\exp(z_y)}{\exp(z_0) + \exp(z_1)}\right) = -z_y + \log(\exp(z_0) + \exp(z_1)) \quad 750$$

where z_0 and z_1 represent logits for biological and artificial chimeric classes, respectively. 751

783 *Training implementation*

784 Training used batch size of 16 sequences with 30 parallel data loading workers. GPU
785 acceleration was employed for efficient processing, with training typically requiring
786 55 hours. Model checkpointing saved the best-performing model based on valida-
787 tion metrics. Configuration management used Hydra [47] to enable reproducible
788 experimentation.

790 *Model evaluation*

791 Performance was monitored using precision, recall, and F1 score on the validation set
792 after each epoch:

$$\begin{aligned} \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \\ \text{F1} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned}$$

799 where TP (true positives) are chimeric reads correctly classified as artificial, TN (true
800 negatives) are biological reads correctly classified as biological, FP (false positives)
801 are biological reads misclassified as artificial, and FN (false negatives) are artificial
802 reads misclassified as biological. Final model selection was based on best validation
803 performance as determined by early stopping.

805 *Model inference and application*

807 *Inference pipeline*

808 To apply ChimeraLM to new WGA sequencing data, the model takes a BAM file as
809 input. Chimeric reads are identified using SA tags and filtered to exclude unmapped,
810 secondary, or supplementary alignments. Each chimeric read sequence is tokenized
811 using the tokenizer (maximum length 32,768 bp, with truncation or padding as
812 needed). The trained model processes sequences in batches, generating two logits
813 $[z_0, z_1]$ for each read corresponding to biological and artificial chimeric classes. Clas-
814 sification is determined by $\hat{y} = \text{argmax}(z_0, z_1)$. ChimeraLM outputs a filtered BAM
815 file containing only reads classified as biological, which can be directly used for
816 downstream analyses including SV calling.

818 *Performance evaluation*

820 *Test set evaluation*

821 Final model performance was evaluated on the held-out test set and the independent
822 MinION Mk1c dataset. Metrics (precision, recall, and F1 score) were computed as
823 described in the training section, where true positives represent chimeric reads cor-
824 rectly classified as artificial and true negatives represent biological reads correctly
825 classified as biological.

Read-length analysis

For every chimeric read (primary alignment with an SA tag) in the WGA PromethION and MinION BAM files, we recorded the full read length (including hard-clipped bases), the number of alignment segments, and the read-coordinate position of every chimeric junction, defined as the boundary between consecutive alignment segments ordered along the read (primary alignment from its CIGAR string; supplementary alignments from the SA tag). A read was considered fully truncated if all of its junctions lay beyond position 32,768. Reads were grouped into length bins (≤ 2 kb, 2–8 kb, 8–16 kb, 16–32 kb, >32 kb) and, for each bin, we computed the fraction of reads classified as artifacts and, for WGA-derived reads of the held-out test set, precision, recall and F1 using the ChimeraLM predictions generated for the full WGA datasets.

SV calling

SVs were called using multiple tools to ensure comprehensive detection. For long-read data (ONT PromethION P2 and MinION Mk1c), we used Sniffles v2.5 [14, 15], DeBreak v1.2 [16], SVIM v2.0.0 [17], and cuteSV v2.1.1 [18]. For short-read data of the PC3 cell line, we used both the CCLE Illumina WGS dataset and the PRJNA361315 Illumina WGS dataset, processed with Manta v1.6.0 [48], DELLY v1.5.0 [49], and SvABA v1.1.0 [50]. All tools were executed with default recommended parameters.

Gold standard SV dataset construction

To evaluate the impact of ChimeraLM on SV detection accuracy, we generated a high-confidence gold-standard SV set from bulk PC3 sequencing data. All SV comparisons and breakpoint corrections were performed using OctopusSV v0.2.3 [51]. Four bulk datasets were integrated: ONT MinION Mk1c, ONT PromethION P2, the CCLE Illumina WGS dataset, and the PRJNA361315 Illumina WGS dataset. SVs were called independently within each dataset, and events supported by at least two SV callers were retained. The remaining calls were then compared across datasets, and SVs observed in at least two of the four datasets were designated as gold-standard events for benchmarking.

SV benchmarking analysis

To assess the impact of ChimeraLM on SV calling accuracy, we compared SV calls from unfiltered WGA data and ChimeraLM-filtered WGA data against two references: (1) the stringent multi-platform gold standard dataset, and (2) platform-matched long-read bulk sequencing data. Benchmarking was performed using Truvari v4.2.2 [52] with default parameters. SVs were considered supported if they matched reference variants within the defined breakpoint tolerance. Validation rates were calculated as the proportion of called SVs supported by the reference. This dual benchmarking strategy quantifies both improvements in detecting high-confidence multi-platform SVs and the retention of platform-specific true variants.

875 **Benchmarking against existing methods**

876 ChimeraLM was compared to existing computational methods for detecting
877 amplification-induced chimeric artifacts: SACRA [25] (GitHub commit 9a2607e), 3rd-
878 ChimeraMiner [13] (GitHub commit 04b5233), and CADECT v1.2 [11]. Both tools
879 were applied to WGA data from PromethION P2 and MinION Mk1c platforms using
880 default parameters as recommended in their documentation. Performance was evalu-
881 ated by measuring the percentage reduction in chimeric reads relative to unprocessed
882 WGA data. Chimeric reads were identified using WGA tag-based alignment criteria
883 (reads with SA tags indicating split alignments), and reduction rates were calculated
884 as the proportion of chimeric reads removed by each method.

886 **Attention weight analysis**

887
888 To investigate ChimeraLM’s interpretability, we analyzed attention weights from
889 the pooling mechanism for representative chimeric reads. Attention weights indicate
890 the relative importance assigned to each sequence position during classification. For
891 selected reads, we extracted per-position attention weights and visualized them along-
892 side read alignments to identify whether the model focuses on mechanistically relevant
893 regions.

894 Chimeric junction positions were identified from alignment data (defined by break-
895 points in SA tags). A region of ± 50 bp surrounding each junction was designated as
896 the junction region. Attention weights within junction region were compared to non-
897 junction regions using the Wilcoxon rank-sum test [53], with statistical significance
898 assessed at $p < 0.001$.

900 **Data visualization**

901
902 Figures were generated using Python with Matplotlib [54] and Seaborn [55].

904 **Computing resources**

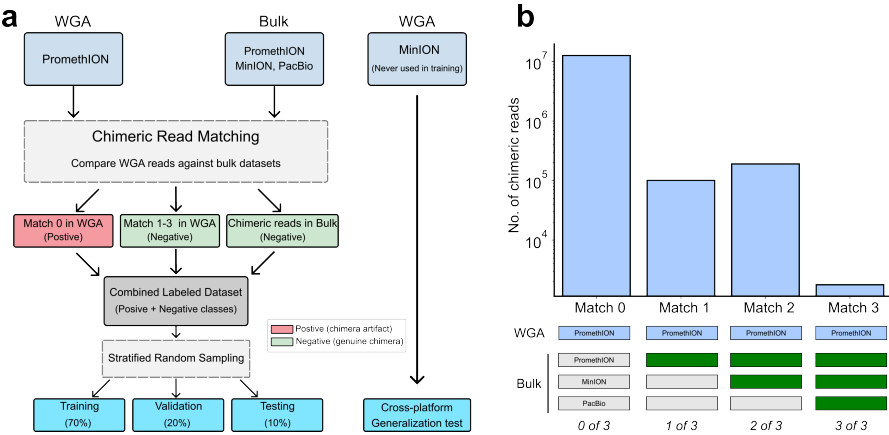
905 Computations were performed on a high performance computing (HPC) server with
906 64-core Intel Xeon Gold 6338 CPU, 256 GB RAM, and two NVIDIA A100 GPUs (80
907 GB memory each).

909 **Supplementary information.**

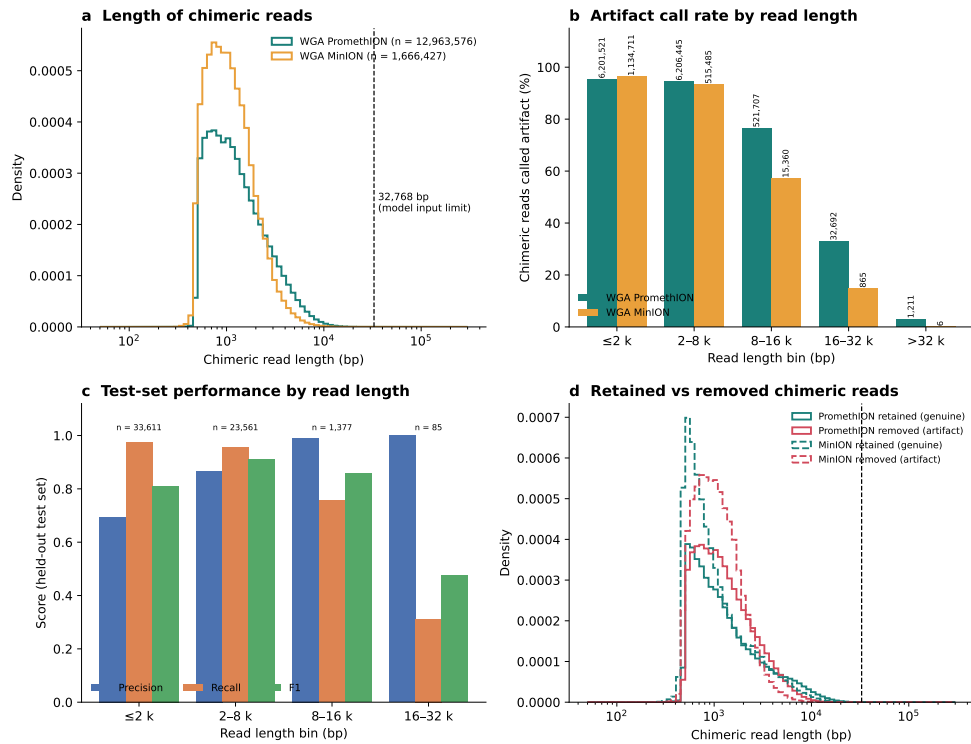
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Extended Data Table 1 Sequencing and alignment statistics of PC3

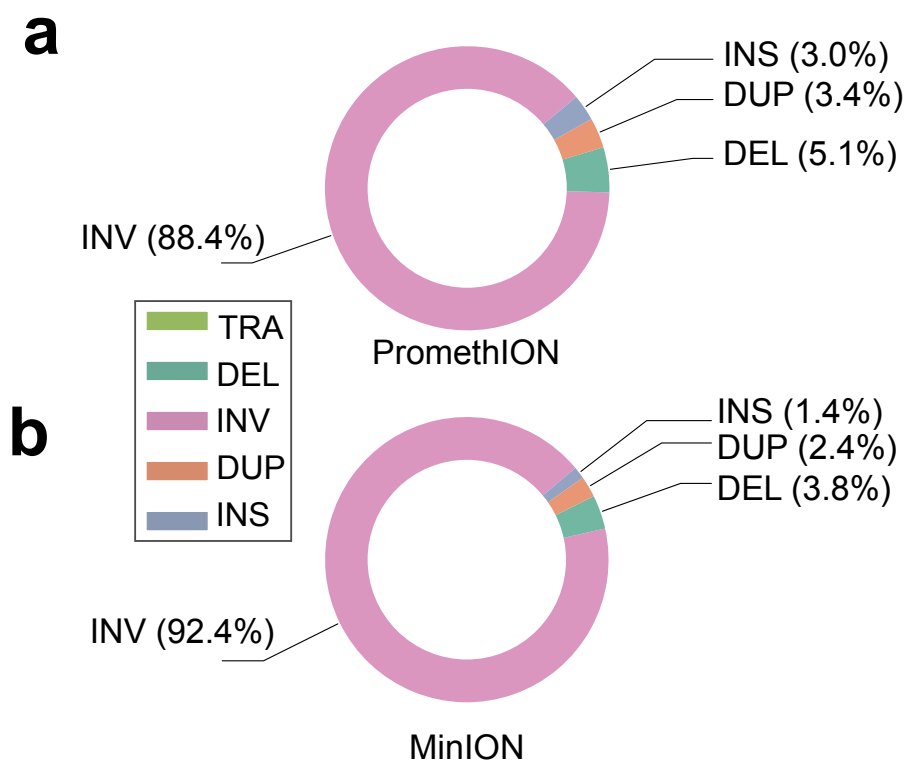
Sample	Platform	Reads ($\times 10^6$)	Total bases (Gb)	Total bases aligned (Gb)	Fraction aligned	Mean length (bp)	Mean quality (Q)	Average identity (%)
WGA	MinION	9.11	14.6	10.4	0.7	1,603	14.3	97.6
WGA	PromethION	44.69	128.2	69.2	0.5	2,869	14.5	96.1
Bulk	MinION	0.97	8.1	7.1	0.9	8,310	17.2	97.3
Bulk	PromethION	8.00	69.9	62.4	0.9	8,732	18.5	97.7



Extended Data Fig. 1 Training dataset construction and bulk-supported labeling strategy. (a) Workflow for generating labeled training data. WGA PromethION data is compared against three independent bulk sequencing datasets (PromethION, MinION, and PacBio). Reads with no bulk matches (Match 0) are labeled artificial; reads matching one or more bulk datasets (Match 1–3) are labeled biological, along with chimeric reads sampled directly from bulk data. The labeled dataset is split into training (70%), validation (20%), and test (10%) sets. The WGA MinION dataset is reserved for independent cross-platform evaluation. (b) Distribution of chimeric read matches. Bar chart shows the number of WGA PromethION chimeric reads (log scale) by bulk dataset matches. Match 0 reads ($\sim 10^7$) lacking bulk validation are classified as artificial; Match 1–3 reads with bulk support are classified as biological. The substantial imbalance reflects high prevalence of WGA-induced artifacts.



Extended Data Fig. 2 Read-length distribution of chimeric reads and length dependence of ChimeraLM classification. (a) Length distribution of chimeric reads (primary alignments carrying an SA tag) in the WGA PromethION ($n = 12,963,576$; median 2,100 bp, N50 4,053 bp, maximum 126,974 bp) and WGA MinION ($n = 1,666,427$; median 1,469 bp, N50 2,333 bp, maximum 42,118 bp) datasets. The dashed line marks the 32,768-bp model input limit, exceeded by 1,211 (0.009%) and 6 reads, respectively; for all but two of these reads at least one chimeric junction lies within the first 32,768 bp. (b) Fraction of chimeric reads classified as WGA-induced artifacts by read-length bin; numbers above bars give the number of reads per bin. (c) Precision, recall and F1 of ChimeraLM on WGA-derived held-out test reads ($n = 58,636$; the 17,876 test reads sampled from bulk data are not shown because they are not part of the WGA dataset) stratified by read length. Bins with fewer than 10 reads are omitted. (d) Length distributions of chimeric reads retained (classified as genuine) and removed (classified as artifact) by ChimeraLM on both platforms (PromethION retained: median 3,087 bp, N50 7,989 bp; removed: median 2,067 bp, N50 3,836 bp; MinION retained: median 2,031 bp, N50 5,560 bp; removed: median 1,458 bp, N50 2,220 bp).



Extended Data Fig. 3 Composition of artifact-supported SVs for PromethION (a) and MinION (b). Donut charts summarize the distribution of SV types including [translocation \(TRA\)](#), [deletion \(DEL\)](#), [inversion \(INV\)](#), [duplication \(DUP\)](#), and [insertion \(INS\)](#) among events supported exclusively by chimera artifacts, representing artificial SVs preferentially removed by ChimeraLM.

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1062

1063 **Declarations**

1064

Author Contributions. YL, QG and RY designed the study. YL and QG performed the analysis. QG performed the experiments. YL and QG designed and implemented the model. YL built the command-line tool and documentation. YL, QG and RY wrote the manuscript. RY supervised this work.

1069

Data Availability. The raw sequencing data generated in this study have been deposited in the NCBI Sequence Read Archive (SRA) under BioProject accession PRJNA1354861. The dataset includes Oxford Nanopore long-read whole-genome sequencing of PC3 prostate cancer cells and MDA-amplified single-cell derivatives. The individual SRA accessions are as follows: PC3 bulk (MinION Mk1C), SRR35904028; PC3 bulk (PromethION P2), SRR35904029; PC3 10-cell WGA (MinION Mk1C), SRR35904026; PC3 10-cell WGA (PromethION P2), SRR35904027. We can access the data at the following link: <https://dataview.ncbi.nlm.nih.gov/object/PRJNA1354861?reviewer=viej6cv6mgbli3n7a9a5k1bsb3>

1078

Code Availability. ChimeraLM, implemented in Python, is open source and available on GitHub (<https://github.com/ylab-hi/ChimeraLM>) under the Apache License, Version 2.0. The package can be installed via PyPI (<https://pypi.org/project/chimeralm>) using pip, with wheel distributions provided for Windows, Linux, and macOS to ensure easy cross-platform installation. An interactive demo is available on Hugging Face (<https://huggingface.co/spaces/yangliz5/ChimeraLM>), allowing users to test ChimeraLM’s functionality without local installation. For large-scale analyses, we recommend using ChimeraLM on systems with GPU acceleration. Detailed system requirements and optimization guidelines are available in the repository’s documentation (<https://ylab-hi.github.io/ChimeraLM/>).

1089

Conflict of interest. The authors declare no competing interests.

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1092 **Acronyms**

1093

CPU central processing unit 13

1094

DEL deletion 8, 23

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dMDA droplet-based MDA 13

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DUP duplication 8, 23

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GLM genomic language model 1, 2, 13

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GPU graphics processing unit 13, 18, 20, 24

1100

HPC high performance computing 20

1101

INS insertion 8, 23

1102

INV inversion	8, 23	1105
LIANTI linear amplification via transposon insertion	13	1106
MALBAC multiple annealing and looping-based amplification cycles	13	1107
MDA multiple displacement amplification	2, 10	1108
MLP multilayer perceptron	4, 5, 16, 17	1109
ONT Oxford Nanopore Technologies	3, 8, 9, 12, 14, 15, 19	1110
PacBio Pacific Biosciences	3, 15	1111
PTA primary template-directed amplification	13	1112
SA supplementary alignment	14, 15, 18–20, 22	1113
SV structural variation	1–5, 8–10, 12, 13, 15, 18, 19, 23	1114
TRA translocation	8, 23	1115
WGA whole genome amplification	1–13, 15, 18–22	1116
WGS whole genome sequencing	8, 9, 19	1117
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